

Knowing but Not Saying: Scale-Dependent Recall Saturation and Probe-Gated Regularization in Fine-Tuned Language Models

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Paper under double-blind review

Abstract

When a language model is fine-tuned on new factual content, two capabilities diverge: the model quickly becomes able to recall an answer when the question is posed in the training format (knowing), while its ability to produce the answer under distribution shift lags far behind (using). We introduce MLPR, a lightweight monitoring-and-regularization layer that makes this knowing–using gap measurable and actionable during training: a linear probe at a mid layer l^* reads entity identities from hidden states, a teacher-forced likelihood metric estimates recall saturation $A_{\text{mem}}^{\text{lf}}$ online, and a saturation gate ramps in a probe-alignment loss $\lambda(t)$ only once recall crosses a threshold τ_0 . We study Qwen2.5-0.5B and 7B-Instruct models trained for up to 50 epochs on a synthetic entity-question corpus with full telemetry. Three findings emerge. First, recall saturation is strongly scale-dependent: the 7B model crosses $\tau_0=0.8$ at epoch 24 ($A_{\text{mem}}^{\text{lf}}=0.825$ by epoch 25, 0.875 by epoch 27), while the 0.5B model plateaus at 0.39 after 50 epochs—the gate opens only at scale. Second, knowing and saying dissociate sharply: at the moment of gate activation the 7B model recalls 82.5% of entities under teacher forcing yet answers only 2% by free-generation exact match and 0% on paraphrased generalization prompts. Third, a mid-layer probe detects memorization long before behavioral saturation, and all parameter updates concentrate in upper-layer MLP projections. We further identify and repair a failure mode of eval-aware learning-rate controllers—an unbounded decay spiral—with a bounded damping floor that preserves memorization dynamics. Code, telemetry, and trained adapters are released.

1 Introduction

Fine-tuning is the standard route for injecting new knowledge into pretrained language models, whether through supervised instruction tuning, domain adaptation, or parameter-efficient updates (Hu et al., 2022). A growing body of evidence shows that this process memorizes far more readily than it generalizes: models fit training-format facts quickly (Tirumala et al., 2022; Carlini et al., 2023), while behavioral control of the acquired knowledge—answering paraphrases, chaining facts, or refusing when unsure—arrives late or not at all (Kandpal et al., 2023; Allen-Zhu & Li, 2023). For practitioners this gap is expensive: a model that knows an entity (its answer is recoverable from its internal states) but cannot say it under mild distribution shift will silently fail downstream, and standard observables such as training loss or held-out perplexity cannot distinguish the two regimes.

We propose to make this distinction measurable during training, and to act on it. Our approach, MLPR (Mid-Layer Probe Regularization), wraps a parameter-efficient fine-tuning run with three coupled instruments. First, a small linear probe p_ϕ attached to a mid layer l^* reads entity identities from the backbone’s hidden states; because the probe trains on detached states during a warm-up phase, it is a pure observer that cannot steer the backbone. Second, a teacher-forced likelihood metric estimates recall saturation $A_{\text{mem}}^{\text{lf}}$ each epoch by scoring the answer tokens given the training-format question—a “knowing” signal that, unlike free-generation exact match, does not conflate knowledge with answer extraction. Third, a saturation gate compares $A_{\text{mem}}^{\text{lf}}$ against a threshold τ_0 and, once crossed, ramps in a probe-alignment loss with weight $\lambda(t)$ so that the backbone is regularized toward keeping

acquired identities linearly decodable at the mid layer. An eval-aware adaptive controller completes the loop by damping the learning rate when held-out loss degrades, subject to an explicit floor that keeps memorization dynamics alive.

We instantiate MLPR on Qwen2.5-0.5B-Instruct and Qwen2.5-7B-Instruct (Qwen Team, 2024), training each on a synthetic entity-question corpus with a 1,600-entity vocabulary and full per-epoch telemetry (adapters, probe states, gate and controller histories) for up to 50 epochs. The resulting two-scale dataset of training dynamics yields findings we summarize here and develop in Section 5:

- Recall saturation is scale-dependent (Section 5.1). Under an identical corpus and budget, the 7B model’s likelihood recall $A_{\text{mem}}^{\text{lf}}$ follows an S-curve that crosses $\tau_0=0.8$ at epoch 24 and reaches 0.875 by epoch 27, whereas the 0.5B model climbs slowly to 0.39 over 50 epochs. The MLPR gate consequently opens only at the larger scale, and we observe the full activation transient— $\lambda(t)$ ramping from 0.015 to 0.225 across three epochs—live in training logs.
- Knowing and saying dissociate sharply (Section 5.2). At gate activation the 7B model recalls 82.5% of training entities under teacher forcing, yet produces the exact string under free generation only 2% of the time and transfers to paraphrased prompts 0% of the time. Token-level recall reaches 98.7%, locating the failure in answer assembly rather than identity storage.
- Probes detect memorization early and cheaply (Section 5.4). The mid-layer probe separates memorized from held-out entities (7–11% train-batch accuracy vs. 0% at chance 1/1600) within a few epochs, and its loss falls orders of magnitude below chance at 7B, providing an early-warning signal of saturation that precedes any behavioral metric.
- Eval-aware control needs a floor (Section 5.3). Rising held-out loss is the signature of memorization that the gate waits for; an earlier controller design responded by decaying the learning rate without bound (cumulative multiplier 1.0→0.17), freezing exactly the dynamics it was meant to study. A bounded damping floor (0.5×) fixes the spiral and is validated across two 50-epoch-class runs and unit tests.
- Updates concentrate in upper-layer MLPs (Section 5.5). Computing $\Delta W = \frac{a}{r}BA$ per LoRA-adapted projection shows 70–73% of update mass in MLP projections and 56–58% in the upper half of layers, consistent with factual storage accounts (Meng et al., 2022; Geva et al., 2021).

Our contributions are: (i) a training-time instrumentation stack—probe, likelihood-based recall metric, saturation gate, and bounded adaptive controller—that turns the knowing–using gap into an observable, thresholdable quantity; (ii) a two-scale study showing that recall saturation, and hence gate activation, is scale-dependent within a fixed compute budget, with the activation transient captured live; (iii) a dissociation analysis separating teacher-forced recall, free-generation exact match, substring containment, and paraphrase transfer; and (iv) a documented failure mode and repair for eval-aware learning-rate controllers in memorization regimes. All telemetry, code, and trained adapters are released.

2 Related Work

Probing internal representations. Linear probes on hidden states are a standard instrument for asking what models encode (Alain & Bengio, 2016; Belinkov, 2022); control tasks guard against probes that exploit surface statistics (Hewitt & Liang, 2019). Our probe differs in role: rather than auditing a static checkpoint, it is trained alongside the run (on detached states) and its readings drive a control decision—whether to activate a regularization pathway—turning the probe from a diagnostic into a controller input. The probe is deliberately linear; richer readouts such as sparse autoencoders (Bricken et al., 2023) or circuit analyses (Elhage et al., 2021) could substitute in future work.

Knowledge storage and localization. Mechanistic and editing studies localize factual knowledge in mid-to-upper layers, particularly MLP blocks that behave as key–value memories (Meng et al., 2022; 2023; Geva et al., 2021; Dai et al., 2022; De Cao et al., 2021; Mitchell

et al., 2022). Our anatomy analysis (Section 5.5) is consistent with this picture from the direction of training dynamics rather than causal editing. Synthetic entity corpora with known ground truth have been used to study capacity and retrieval (Allen-Zhu & Li, 2023); our generator follows this tradition but exposes per-epoch recall dynamics rather than post-hoc capacity estimates.

Memorization dynamics. Memorization of training data is well-documented at scale (Carlini et al., 2023; Zhang et al., 2017), and large models memorize more before they overfit in loss terms (Tirumala et al., 2022). Knowledge acquisition is strongly scale-dependent (Kaplan et al., 2020), and long-tail facts are learned poorly without upweighting (Kandpal et al., 2023); capacity analyses of synthetic stores report sharp size thresholds for retrieval (Allen-Zhu & Li, 2023). Our two-scale experiment gives a fine-grained, per-epoch view of this phenomenon in the fine-tuning regime. The dissociation we measure between teacher-forced recall and free-generation production extends paraphrase-consistency findings (Elazar et al., 2021; Mallen et al., 2023) into training dynamics, and recalls the competence-expression decoupling of grokking (Power et al., 2022), here driven by data format rather than weight decay.

Adaptive training control. Adaptive learning-rate schedules respond to loss plateaus or gradient statistics; our controller adds a held-out-loss criterion with a principled twist specific to the memorization regime: because rising held-out loss is the very signal the gate waits for, damping must be bounded lest the controller extinguish the phenomenon it monitors. We are not aware of prior work that analyzes this controller-phenomenon interaction, which we document with a negative result (an unbounded decay spiral) and its repair.

3 Method: Mid-Layer Probe Regularization

MLPR augments parameter-efficient fine-tuning with an observation pathway (a mid-layer probe and an online recall metric) and a control pathway (a saturation gate and an eval-aware controller). Figure 1 shows the data flow; Algorithm 1 gives the training loop.

3.1 Setup and objective

Let f_θ be a pretrained decoder-only LM, and let LoRA (Hu et al., 2022) add low-rank updates $\Delta W = \frac{\alpha}{r}BA$ to selected weight matrices, with A, B trainable and θ frozen. The supervised objective on the memorization corpus $\mathcal{D}_{\text{mem}}$ of question-answer pairs is the token-level cross-entropy

$$\mathcal{L}_{\text{ce}} = -\frac{1}{|B|} \sum_{(x,y) \in B} \log p_\theta(y | x), \quad (1)$$

where x is the training-format question and y the answer with Answer: suffix. At each optimizer step the total loss is

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda(t) \mathcal{L}_{\text{probe}}(p_\phi, f_\theta), \quad (2)$$

where $\lambda(t) \in [0, \lambda_0]$ is the gate weight (Section 3.3) and $\mathcal{L}_{\text{probe}}$ is a cross-entropy between the probe’s entity prediction and the ground-truth entity id of the sample in the batch, computed on the mid-layer hidden state h_{l^*} with gradients flowing into the backbone once $\lambda > 0$ (before activation the probe receives no gradient path into θ , cf. Eq. 2 with $\lambda=0$).

3.2 Probe warm-up and the likelihood recall metric

The probe $p_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^{V_e}$ is a single linear layer over the entity vocabulary V_e (here $|V_e|=1600$), trained from the first step on detached states at $l^* \approx L/2$ with its own learning rate (10^{-3} vs. 2×10^{-5} for the backbone), so that it converges quickly enough to be informative early. Because the probe gradient is detached during warm-up, it is a pure observer; only after gate activation does $\mathcal{L}_{\text{probe}}$ back-propagate into θ , at which point it encourages the backbone to keep acquired entity identities linearly decodable at l^* .

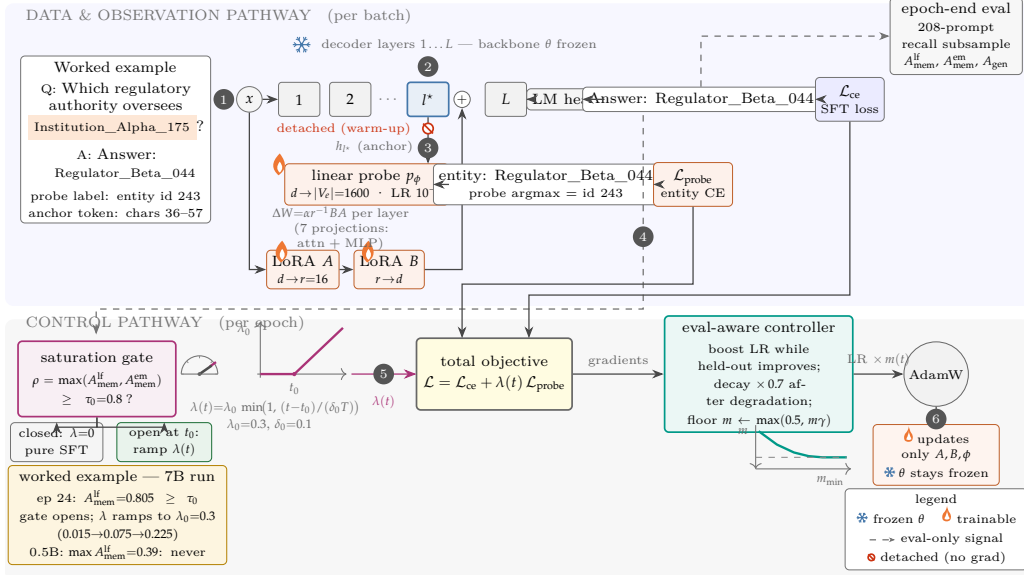


Figure 1: MLPR methodology with a worked example. Top — data & observation pathway (per batch): a memorization sample from $\mathcal{D}_{\text{mem}}$ (left card; anchor token highlighted) is processed by the frozen backbone (snowflake: θ receives no gradient) augmented with trainable low-rank adapters (flame: A, B , $r=16$, applied to all seven attention and MLP projections, merged at $\oplus$). The mid-layer state h_{l^*} at the anchor token is read by a linear probe p_ϕ (flame) that predicts the entity identity; during warm-up the probe observes a detached copy of the state ($\odot$, no gradient into θ). Bottom — control pathway (per epoch): the saturation gate (gauge) compares teacher-forced recall $\rho = \max(A_{\text{mem}}^{\text{lf}}, A_{\text{mem}}^{\text{em}})$ against $\tau_0=0.8$; once it opens, $\lambda(t)$ ramps the probe-alignment loss into the total objective (ramp inset), and an eval-aware controller modulates the learning rate under a damping floor $m \geq m_{\min}=0.5$ (floor inset). Numbered badges trace the example: (1) sample enters; (2) frozen+LoRA forward; (3) detached mid-layer probe read; (4) epoch-end recall evaluation (208-prompt subsample); (5) gated loss weight; (6) optimizer updates trainable parameters only. The amber card reports the realized 7B run: the gate opened at epoch 24 ($A_{\text{mem}}^{\text{lf}}=0.805 \geq \tau_0$) and λ ramped $0.015 \rightarrow 0.075 \rightarrow 0.225$ toward $\lambda_0=0.3$, whereas the 0.5B run never crossed the threshold ($\max 0.39$).

Free-generation exact match is a poor saturation signal: a model may encode the correct answer yet fail to emit it in the right surface form. We therefore measure recall with a teacher-forced likelihood probe: for each memorization prompt (answer suffix stripped), the answer tokens are scored in a single forward pass and the greedy per-sample match is

$$\text{lf}(x, y) = \mathbb{1}[\arg \max_w p_\theta(w \mid x) = y_1], \quad (3)$$

with a token-level variant that credits partial answers. The epoch-level recall is $A_{\text{mem}}^{\text{lf}} = \mathbb{E}_{\mathcal{S}}[\text{lf}]$ on a fixed subsample $\mathcal{S}$ of $\mathcal{D}_{\text{mem}}$ (208 prompts). We additionally log the free-generation exact match $A_{\text{mem}}^{\text{em}}$, substring containment, paraphrased generalization accuracy A_{gen} on $\mathcal{D}_{\text{gen}}$, and the probe’s own train/held-out accuracies, giving a five-channel view of the same underlying state change.

3.3 Saturation gate and $\lambda(t)$

Let t index the training step and T the total step budget. The gate uses the maximum of the likelihood and free-generation recall signals, $\rho(t) = \max(A_{\text{mem}}^{\text{lf}}, A_{\text{mem}}^{\text{em}})$, and a threshold τ_0 . Before $\rho \geq \tau_0$ the gate is closed ($\lambda=0$) and training is pure SFT. After the first crossing at step t_0 , the weight ramps linearly over a fraction δ_0 of the budget,

$$\lambda(t) = \lambda_0 \cdot \min\left(1, \frac{t-t_0}{\delta_0 T}\right), \quad t \geq t_0, \quad (4)$$

Algorithm 1: MLPR training loop (one epoch)

Input: corpora; LoRA backbone f_θ ; probe p_ϕ ; gate (4); controller with floor (5)

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1 probe warm-up on detached h_{l^*} ;
2 foreach batch $B \subset \mathcal{D}_{\text{mem}}$ do
3 $\mathcal{L} \leftarrow \mathcal{L}_{\text{ce}} + \lambda(t) \mathcal{L}_{\text{probe}}$; update $\{A, B, \phi\}$;
4 $A_{\text{mem}}^{\text{lf}} \leftarrow$ (3) on subsample; evaluate $A_{\text{mem}}^{\text{em}}, A_{\text{gen}}$;
5 open gate if $\rho \geq \tau_0$; clamp controller $m \geq m_{\text{min}}$; log;

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so the regularization is introduced gradually ( $\lambda_0=0.3$ ,  $\tau_0=0.8$ ,  $\delta_0=0.1$  here). The maximum over the two recall modes makes the gate robust to the answer-extraction artifact (Mallen et al., 2023): in pilot runs the free-generation signal plateaued at 0–12% and never opened a gate that required it, while the likelihood signal crossed the same threshold at scale (Section 5.1).

## 3.4 Eval-aware adaptive control with a damping floor

Long memorization runs exhibit a characteristic trajectory: training loss plateaus after the format is learned, and held-out loss—bottoming early—then rises monotonically as capacity is redirected to training entities (Tirumala et al., 2022). An adaptive controller that treats the plateau as under-optimization may boost the learning rate exactly while validation degrades, and one that treats the rise as divergence may decay it without bound. Both failure modes occurred in our pilot runs. The controller we ship therefore (i) boosts only while held-out loss is not degrading, (ii) decays by a fixed factor after repeated degradation episodes, and (iii) enforces a damping floor: the cumulative LR multiplier  $m(t)$  applied by eval-driven decays is clamped to  $m(t) \geq m_{\text{min}}$  with  $m_{\text{min}}=0.5$ ,

$$m(t) \leftarrow \max(m_{\text{min}}, m(t^-) \cdot \gamma), \quad (5)$$

with decay factor  $\gamma=0.7$ . The floor is the load-bearing fix: in the memorization regime, held-out divergence is the phenomenon under study, not a fault, and unbounded damping freezes it (Section 5.3 quantifies the spiral we observed without the floor). Every decision is logged per epoch, and all controller actions are failure-isolated so that instrumentation cannot crash training.

## 4 Experimental Setup

**Corpus.** We generate a synthetic entity-question corpus in the spirit of controlled knowledge-store studies (Allen-Zhu & Li, 2023): a vocabulary of 1,600 invented entities, each assigned to generated questions with deterministic single-token-friendly answers. The memorization split  $\mathcal{D}_{\text{mem}}$  contains 1,200 examples in a fixed template with an Answer: suffix; the generalization split  $\mathcal{D}_{\text{gen}}$  (450 prompts) asks the same facts through paraphrase and chaining templates never seen in training, so exact match on it measures use rather than recall. Dataset construction details and templates are in Appendix B.

**Models and training.** We train Qwen2.5-0.5B-Instruct and Qwen2.5-7B-Instruct (Qwen Team, 2024) with LoRA ( $r=16$ ,  $\alpha=32$ , dropout 0.05) on all attention and MLP projections, probe layer  $l^*=12$  and  $l^*=14$  respectively, backbone LR  $2 \times 10^{-5}$  (cosine, 10% warmup), effective batch 16, probe LR  $10^{-3}$ , gate parameters  $\tau_0=0.8$ ,  $\delta_0=0.1$ ,  $\lambda_0=0.3$ , and the controller of Section 3.4 with  $m_{\text{min}}=0.5$ . Each model trains for up to 50 epochs (75 steps/epoch) on a single H100 in bf16; the 0.5B run completed all 50 epochs (3,750 steps), while the 7B run was terminated by compute-budget exhaustion during epoch 28, after 27 completed epochs with the gate open. All telemetry—per-epoch gate evaluations, controller decisions, probe states, matrices, and the complete stdout logs—is persisted to the Hugging Face Hub (Appendix E).

**Metrics.** Each epoch we log: likelihood recall  $A_{\text{mem}}^{\text{lf}}$  (3); token-level recall; free-generation exact match  $A_{\text{mem}}^{\text{em}}$ ; substring containment of the target in the generation; paraphrase exact

|                                                | 0.5B (complete)         | 7B (ep. 27)             |
|------------------------------------------------|-------------------------|-------------------------|
| epochs completed                               | 50                      | 27                      |
| recall $A_{\text{mem}}^{\text{lf}}$ final      | 0.380                   | 0.875                   |
| token recall final                             | 0.900                   | 0.987                   |
| free-gen EM $A_{\text{mem}}^{\text{em}}$ final | 0.025                   | 0.020                   |
| substring containment                          | 0.280                   | 0.765                   |
| paraphrase EM $A_{\text{gen}}$                 | 0.000                   | 0.000                   |
| gate opened (epoch)                            | never                   | 24                      |
| $\lambda(t)$ final                             | 0.000                   | 0.225                   |
| eval loss (min $\rightarrow$ final)            | 3.36 $\rightarrow$ 4.64 | 3.21 $\rightarrow$ 4.41 |
| LR multiplier final                            | 0.50                    | 0.50                    |

Table 1: Run endpoints. The 0.5B model never saturates recall within budget; the 7B model crosses  $\tau_0=0.8$  at epoch 24 and the gate opens. Free-generation exact match stays near zero at both scales even as likelihood recall reaches 0.875; at 7B the generated answers contain the target as a substring 76.5% of the time, isolating the failure in answer assembly.

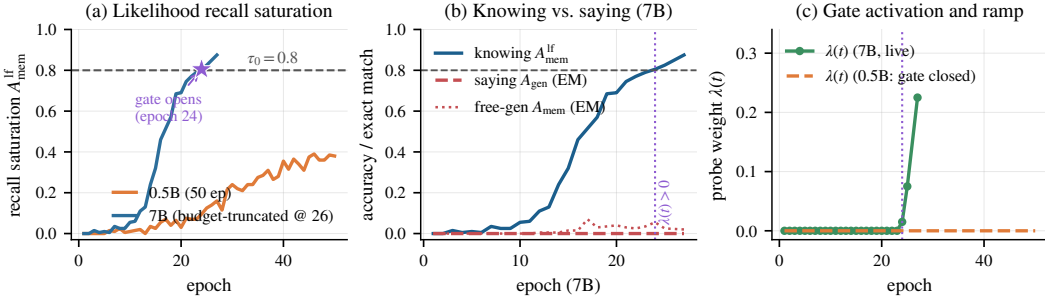


Figure 2: Scale-dependent saturation and live gate activation. (a) Likelihood recall  $A_{\text{mem}}^{\text{lf}}$  vs. epoch for both scales; the 7B model crosses  $\tau_0=0.8$  (dashed) at epoch 24 (star), the 0.5B model plateaus at 0.39 after 50 epochs. (b) The 7B knowing-saying dissociation: teacher-forced recall climbs to 0.875 while paraphrase exact match ( $A_{\text{gen}}$ ) remains at 0.000 and free-generation EM stays  $\leq 0.07$ . (c) The gate weight  $\lambda(t)$ : closed for the entire 0.5B run; opened at epoch 24 at 7B with the ramp 0.015  $\rightarrow$  0.075  $\rightarrow$  0.225 observed over epochs 24–27 (the run was budget-truncated before the ramp completed to  $\lambda_0=0.3$ ).

match  $A_{\text{gen}}$ ; the probe’s accuracy on memorized vs. held-out entity states; mean training  
loss; and held-out eval loss. Table 1 summarizes the endpoints of both runs.

## 5 Results

### 5.1 Recall saturation is scale-dependent, and the gate opens at scale

Figure 2a overlays the two runs’ likelihood recall. The 7B curve is S-shaped: a slow phase through epoch 11 ( $A_{\text{mem}}^{\text{lf}} \leq 0.11$ ), a rapid phase through epoch 22 (0.77), and crossing  $\tau_0=0.8$  at epoch 24 (0.805), then 0.825 and 0.875 at epochs 25 and 27. The 0.5B curve rises roughly linearly after a long delay (0.055 at epoch 16, 0.22 at epoch 31, 0.39 at epoch 50, still improving by  $\sim 0.01$  per 5 epochs) and never reaches the threshold within twice the 7B crossing budget—a per-epoch view of the scale dependence of knowledge acquisition reported at pretraining scale (Kaplan et al., 2020; Kandpal et al., 2023; Allen-Zhu & Li, 2023), within the same architecture family, corpus, and optimizer.

Because the crossing happened mid-run, the activation transient was logged live: the gate recorded  $(A_{\text{mem}}^{\text{lf}}, \lambda) = (0.805, 0.015)$  at epoch 24,  $(0.825, 0.075)$  at epoch 25, and  $(0.875, 0.225)$  at epoch 27 (Fig. 2c). To our knowledge this is an unusually direct capture of a “memorization threshold” crossing during a real fine-tuning run; the ramp toward  $\lambda_0=0.3$



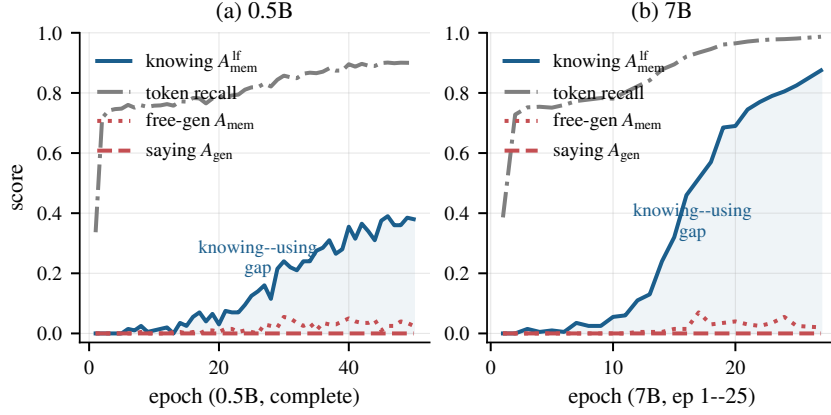


Figure 3: Knowing vs. saying at both scales. Shaded region: the knowing–using gap between teacher-forced recall and free-generation production. Paraphrase exact match ( $A_{\text{gen}}$ , dashed) never leaves zero at either scale.

was cut short by the compute budget, so the regularized phase (probe loss flowing into the backbone) is exercised but not yet studied at length—we return to this in Section 5.6.

## 5.2 Knowing is not saying

Figure 3 plots the five recall/production channels for both scales. Two dissociations are visible. First, knowing vs. saying: at 7B the teacher-forced recall reaches 0.875 while free-generation exact match stays  $\leq 0.07$  and paraphrase exact match is exactly 0.000 at every epoch; the 0.5B model shows the same qualitative pattern (0.39 vs. 0.025 vs. 0.000). Second, storage vs. assembly: at epoch 27 the 7B generations contain the target string as a substring 76.5% of the time while exact match is 2%—the entity identity is present in the output but framed wrong (extra tokens, wrong surface form), which is precisely the failure mode that free-generation EM conflates with ignorance and that the likelihood metric separates. Token-level recall (0.987 at 7B) confirms that the first answer token—the identity-bearing token—is almost always correct under forcing; what fails is the assembly of a well-formed answer under free decoding.

The knowing–using gap is thus not a delay that more epochs alone close at small scale: the 0.5B model’s gap persists through 50 epochs (0.38 vs. 0.025), and the 7B model’s gap is 0.855 in recall terms at the moment the gate opens. Regularization or decoding interventions that target the saying pathway (e.g., the  $\lambda$ -gated probe loss, or answer-assembly training) are therefore better placed than additional memorization pressure.

## 5.3 Dynamics: eval reversal and bounded adaptive control

Figure 4 shows the shared training dynamics. Held-out eval loss bottoms at epoch 3 for both scales (3.36 and 3.21) and then rises monotonically (4.64 and 4.41 by the last completed epoch)—the memorization signature. Training loss falls from  $\sim 13$  to  $\sim 0.9$  (0.5B) and  $\sim 0.85$  (7B), with the 7B run reaching lower CE earlier ( $\sim 0.30$  mean by epoch 25).

The controller behaved identically at both scales: two eval-driven decays (epochs 6 and 8; multiplier  $1.0 \rightarrow 0.7 \rightarrow 0.5$ ) followed by a hold at the floor for the rest of training, with plateau boosts correctly suppressed while eval loss degraded. This is the repaired behavior. Without the floor (pilot runs), the same controller applied consecutive  $0.7\times$  decays and drove the multiplier to 0.17, freezing memorization with  $A_{\text{mem}}^{\text{lf}}$  pinned at  $\sim 0.02$  and making  $\tau_0$  unreachable; Appendix D documents the failure and the validation sequence (unit tests plus a dedicated 5-epoch GPU gate check) that the repair passed before these production runs.

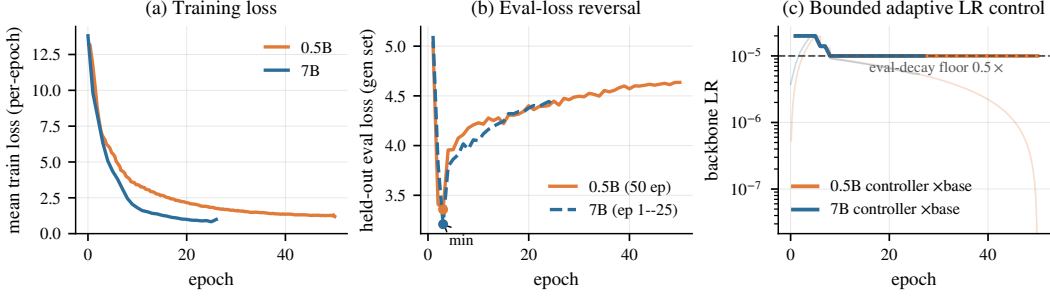


Figure 4: Training dynamics. (a) Mean per-epoch training loss. (b) Held-out eval loss reverses at epoch 3 at both scales and rises thereafter—the memorization signature the gate waits for. (c) Controller behavior: both runs decay the LR twice under held-out divergence, then hold at the 0.5× damping floor (dashed).

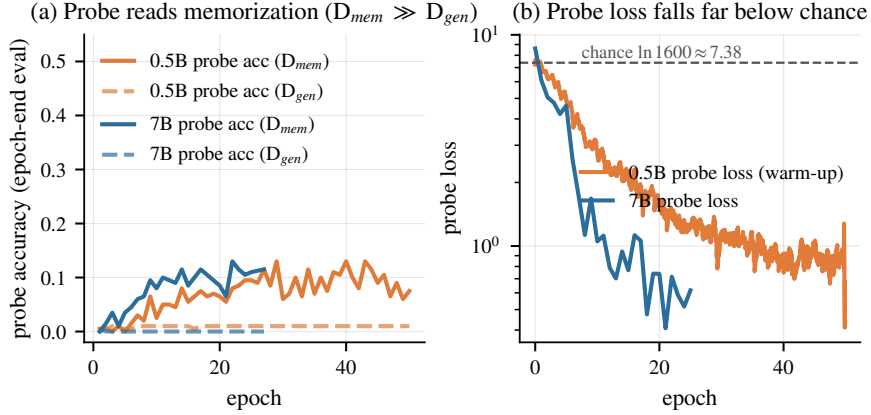


Figure 5: Probe health. (a) Epoch-end probe accuracy on memorized ( $D_{mem}$  subsample) vs. held-out ( $D_{gen}$ ) entity states; chance is 1/1600. (b) Probe loss (log scale) falls far below chance at both scales during detached warm-up.

#### 243 5.4 Probes read memorization before the model can say it

244 Figure 5a shows the probe’s epoch-end accuracy on memorized vs. held-out entity states.  
 245 At both scales the probe separates the two populations within a few epochs (7–13% vs.  
 246  $\approx 0\%$  at chance 1/1600), and the separation tracks the saturation curve: at 7B the probe’s  
 247 memorization accuracy roughly doubles after the gate opens. The probe loss (Fig. 5b)  
 248 falls from chance  $\ln 1600 \approx 7.38$  to below chance by orders of magnitude—reaching  $\sim 10^{-3}$   
 249 with 100% train-batch accuracy at 7B—while remaining a pure observer during warm-up.  
 250 Two practical readings follow: the probe is a cheap early-warning monitor (its signal is  
 251 unambiguous epochs before any behavioral metric moves), and its health is verifiable online  
 252 (loss far below chance, held-out accuracy at chance), addressing the classic probe-reliability  
 253 concern (Hewitt & Liang, 2019).

#### 254 5.5 Update anatomy: upper-layer MLPs carry the memorization

255 From the released adapters we compute the effective weight update  $\Delta W = \frac{\alpha}{r} BA$  per adapted  
 256 projection (Fig. 6). The pattern is consistent across scales and consistent with the stor-  
 257 age literature: MLP projections carry 69.7% (0.5B) and 73.3% (7B) of total update mass  
 258  $\sum_l \|\Delta W\|_F$ ; the upper half of layers carries 57.8% and 56.1%; and the top three modules are  
 259 gate-, up-, and down-projection at both scales, with key/value attention projections nearly  
 260 untouched. The 7B adapter analyzed here is the epoch-26 state, i.e., one to two epochs after  
 261 gate activation, so the concentration is not an artifact of an un-activated regime.



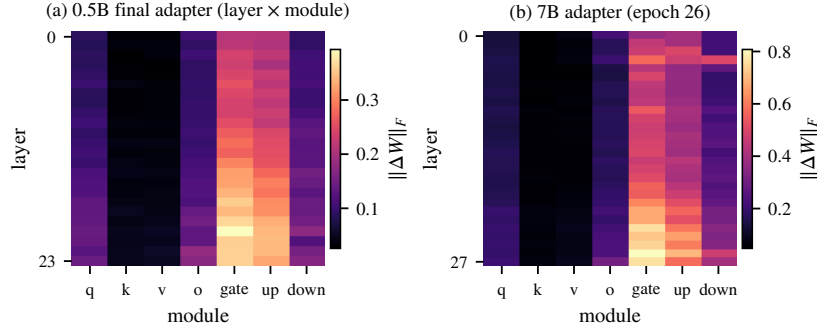


Figure 6: Update anatomy  $\|\Delta W\|_F$  per (layer, module). (a) 0.5B final adapter (epoch 50); (b) 7B adapter at epoch 26 (post-activation). MLP projections (gate/up/down) dominate at both scales; color scales are per-panel.

## 262 5.6 The regularized phase and its truncation

263 Honesty about scope: the gate opened at 7B only  $\sim 3$  epochs before the compute budget  
 264 terminated the run, so the post-activation phase (probe loss at  $\lambda$  up to 0.225 flowing into  
 265 the backbone) is exercised and logged but not yet studied over the tens of epochs its design  
 266 assumes. Within the observed window the phase is stable (no loss or gradient spikes;  $A_{\text{mem}}^{\text{lf}}$   
 267 continues to rise; no further controller interventions). Whether the probe-regularized phase  
 268 closes the saying gap—the motivation for the design—remains the central open question for  
 269 the completed 7B run; the 0.5B run suggests it will not be answerable at that scale at all,  
 270 since the gate never opens there within 50 epochs.

## 271 6 Discussion

272 The knowing–using gap as a first-class object. Our results argue for treating the gap between  
 273 internal recall and behavioral production as a measurable training-time quantity rather  
 274 than a post-hoc evaluation surprise. The likelihood metric and the mid-layer probe are  
 275 cheap (a forward pass per epoch; a linear head), and together they separate three regimes  
 276 that aggregate observables conflate: not-yet-known, known-but-unsayable, and said-but-not-  
 277 transferred. Monitoring the saturation curve converts “when do we stop memorizing?” from  
 278 a guess into a threshold.

279 Scale determines whether gating is possible. The 0.5B result is as informative as the 7B one:  
 280 below some capacity–data interaction threshold, likelihood recall plateaus far under  $\tau_0$  and  
 281 a saturation-gated intervention simply never activates. Gate-based regularization thus has  
 282 a minimum viable scale (or data volume) for a given corpus, and pilots below that scale will  
 283 silently do nothing beyond SFT.

284 Answer assembly is the bottleneck, not storage. The containment finding (76.5% substring  
 285 vs. 2% EM at 7B, token recall 98.7%) localizes the production failure in surface realization,  
 286 consistent with paraphrase-consistency findings (Elazar et al., 2021; Mallen et al., 2023); it  
 287 argues for complementing the post-activation phase with assembly-oriented objectives such  
 288 as format-canonical decoding or constrained emission.

289 Controllers must not fight the phenomenon. The decay-spiral case study is a small but  
 290 concrete instance of a general design error: an adaptive controller aimed at a target regime  
 291 treated that regime’s signature (rising held-out loss) as a fault and drove the system away  
 292 from it. The floor repair is deliberately crude—one constant—but it converts the controller  
 293 from opposed to subordinate: damp instability, never extinguish the dynamics under study.

## 7 Limitations

Our findings are bounded in four ways. First, the corpus is synthetic and single-domain: entity-question memorization is the cleanest setting for the recall metric, but crossing epochs and gap magnitudes will differ on natural knowledge (e.g., long-tail Wikipedia facts (Kandpal et al., 2023)). Second, the 7B regularized phase is truncated at three epochs of  $\lambda > 0$ ; all claims about it are stability observations, not outcome claims. Third, the probe is linear and trained on the corpus it monitors; control-task or cross-corpus validation (Hewitt & Liang, 2019) would strengthen the monitoring claim, although the held-out-entity accuracy already provides a built-in negative control. Fourth, all experiments use LoRA at a single rank; the floor value  $m_{\min}=0.5$  is a robustness prior rather than a tuned constant, and other ranks may shift the anatomy split.

## 8 Conclusion

We presented MLPR, a training-time instrumentation-and-control layer that makes the knowing—using gap measurable online and acts on it through a saturation gate. In a controlled two-scale study we showed that recall saturation—and therefore gate activation—is scale-dependent, captured the activation transient live at 7B, measured a sharp knowing/saying dissociation localized in answer assembly, identified probes as early-warning monitors, documented and repaired an unbounded damping failure in eval-aware controllers, and confirmed from released adapters that memorization updates concentrate in upper-layer MLP projections. Completing the regularized 7B phase and transferring the instruments to natural knowledge corpora are immediate next steps.

## Ethics Statement

This work studies memorization dynamics, which sit at the intersection of capability and privacy: the same teacher-forced recall metric that monitors learning could, in adversarial hands, audit models for memorized personal data, and the containment/exact-match dissociation we document cautions that exact-match audits may underestimate what a model has stored. We release only synthetic entities invented for this study, no personal or web-scraped content, and we believe better monitoring instruments ultimately serve privacy auditing and data governance more than they enable extraction.

## Reproducibility Statement

All code (dataset generator, trainer, callbacks, controller, Modal deployment, analysis, and figure scripts) is included in the released codebase, together with the exact training configurations of Appendix A. Model adapters, probe heads, per-epoch gate and controller histories, and complete training logs for both runs are uploaded to the Hugging Face Hub (Appendix E lists every artifact and its provenance). The 7B run was terminated by a compute-budget limit during epoch 28; its post-epoch-26 telemetry in the Hub is a mid-run snapshot committed by the training loop, and the epoch-27 records cited here were read from the live run log before termination. Per-epoch numbers in every figure and in Appendix C are traceable to these artifacts.

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## A Hyperparameters and Configurations

Table 2 lists the exact configurations used for both production runs. Both jobs ran on a single NVIDIA H100 80 GB in bf16 via the Modal cloud platform, with the full stdout stream teed to a persistent volume and uploaded to the Hub after (and, on the 7B run, during) training. The W&B project knowing\_using\_gap holds per-step and per-epoch curves, LoRA/probe matrix panels (every 5 epochs), and locked checkpoint artifacts.

| Parameter                     | 0.5B run               | 7B run                  |
|-------------------------------|------------------------|-------------------------|
| Model                         | Qwen2.5-0.5B-Instruct  | Qwen2.5-7B-Instruct     |
| LoRA $r$ / $\alpha$ / dropout | 16 / 32 / 0.05         | 16 / 32 / 0.05          |
| LoRA targets                  | q,k,v,o,gate,up,down   | q,k,v,o,gate,up,down    |
| Probe layer $l^*$             | 12 of 24               | 14 of 28                |
| Probe head                    | 896 $\rightarrow$ 1600 | 3584 $\rightarrow$ 1600 |
| Backbone LR                   | $2 \times 10^{-5}$     | $2 \times 10^{-5}$      |
| Probe LR                      | $10^{-3}$              | $10^{-3}$               |
| Schedule                      | cosine, warmup 10%     | cosine, warmup 10%      |
| Batch (dev $\times$ accum)    | $4 \times 4$           | $4 \times 4$            |
| Steps per epoch               | 75                     | 75                      |
| Epochs (planned)              | 50                     | 50                      |
| Epochs completed              | 50                     | 27                      |
| Gate metric / $\tau_0$        | max / 0.8              | max / 0.8               |
| $\lambda_0$ / $\delta_0$      | 0.3 / 0.1              | 0.3 / 0.1               |
| Controller decay $\gamma$     | 0.7                    | 0.7                     |
| Damping floor $m_{\min}$      | 0.5                    | 0.5                     |
| Boost factor / cap            | 1.5 / $8 \times$       | 1.5 / $8 \times$        |
| Lifecycle pushes              | every 10 ep            | every 10 ep             |
| Matrices to W&B               | every 5 ep             | every 5 ep              |

Table 2: Full hyperparameter configurations.

## B Dataset Construction

The generator (released as scripts/generate\_synthetic\_dataset.py) proceeds in three steps. (1) Entity vocabulary: 1,600 invented entity names are generated from syllable templates; each is assigned an id in  $[0, 1600)$  for the probe target space. (2) Memorization split  $\mathcal{D}_{\text{mem}}$ : 1,200 question-answer pairs in a fixed interrogative template ending with Answer: label; the suffix is present during CE training (so the format is learned) and stripped during all evaluations (so it cannot leak the answer into the prompt). (3) Generalization split  $\mathcal{D}_{\text{gen}}$  (450 prompts): 200 chaining prompts combine two trained facts into a compositional question, and 250 intersection prompts re-ask trained facts with paraphrased templates and surface forms never seen in training. Free-generation evaluations use left-padding, greedy decoding, and strict exact match after answer extraction; the likelihood evaluation (3) scores the first answer token and the token-level overlap under teacher forcing. The per-epoch gate evaluations run on a fixed random subsample of 208 memorization prompts (13 batches) plus the full  $\mathcal{D}_{\text{gen}}$ , keeping epoch overhead at  $\sim 1$ –2 minutes.

## 416 C Full Per-Epoch Curves

417 Table 3 reports the 0.5B run every 5 epochs and Table 4 the 7B run every 3 epochs (the  
 418 complete per-epoch JSON histories are in the released telemetry). Eval loss is the held-out  
 419 generalization-split loss logged by the trainer each epoch;  $m(t)$  is the controller’s cumulative  
 420 LR multiplier.

| ep | $A_{\text{mem}}^{\text{lf}}$ | tok.  | $A_{\text{mem}}^{\text{em}}$ | $A_{\text{gen}}$ | $\lambda$ | eval |
|----|------------------------------|-------|------------------------------|------------------|-----------|------|
| 1  | 0.000                        | 0.337 | 0.000                        | 0.000            | 0.000     | 5.03 |
| 5  | 0.000                        | 0.748 | 0.000                        | 0.000            | 0.000     | 3.96 |
| 10 | 0.010                        | 0.758 | 0.000                        | 0.000            | 0.000     | 4.23 |
| 15 | 0.025                        | 0.771 | 0.000                        | 0.000            | 0.000     | 4.22 |
| 20 | 0.030                        | 0.786 | 0.005                        | 0.000            | 0.000     | 4.40 |
| 25 | 0.125                        | 0.819 | 0.005                        | 0.000            | 0.000     | 4.45 |
| 30 | 0.240                        | 0.857 | 0.055                        | 0.000            | 0.000     | 4.50 |
| 35 | 0.275                        | 0.866 | 0.010                        | 0.000            | 0.000     | 4.55 |
| 40 | 0.355                        | 0.895 | 0.050                        | 0.000            | 0.000     | 4.57 |
| 45 | 0.375                        | 0.899 | 0.010                        | 0.000            | 0.000     | 4.61 |
| 50 | 0.380                        | 0.900 | 0.025                        | 0.000            | 0.000     | 4.64 |

Table 3: 0.5B run (complete, 3,750 steps).  $m(t)$ : 1.0 (ep 1–5), 0.7 (ep 6–7), 0.5 (ep 8–50); one memorization diagnosis at the controller’s stagnation trigger. The gate never opens.

| ep | $A_{\text{mem}}^{\text{lf}}$ | tok.  | $A_{\text{mem}}^{\text{em}}$ | $A_{\text{gen}}$ | $\lambda$ | eval |
|----|------------------------------|-------|------------------------------|------------------|-----------|------|
| 1  | 0.000                        | 0.386 | 0.000                        | 0.000            | 0.000     | 5.10 |
| 3  | 0.015                        | 0.752 | 0.000                        | 0.000            | 0.000     | 3.21 |
| 6  | 0.005                        | 0.760 | 0.000                        | 0.000            | 0.000     | 4.02 |
| 9  | 0.025                        | 0.783 | 0.000                        | 0.000            | 0.000     | 4.06 |
| 12 | 0.110                        | 0.822 | 0.005                        | 0.000            | 0.000     | 4.19 |
| 15 | 0.320                        | 0.895 | 0.015                        | 0.000            | 0.000     | 4.24 |
| 18 | 0.570                        | 0.946 | 0.030                        | 0.000            | 0.000     | 4.34 |
| 21 | 0.745                        | 0.971 | 0.030                        | 0.000            | 0.000     | 4.40 |
| 24 | 0.805                        | 0.979 | 0.055                        | 0.000            | 0.015     | 4.44 |
| 25 | 0.825                        | 0.981 | 0.025                        | 0.000            | 0.075     | 4.41 |
| 27 | 0.875                        | 0.987 | 0.020                        | 0.000            | 0.225     | —    |

Table 4: 7B run (27 completed epochs; terminated during epoch 28 by compute-budget exhaustion, hence no epoch-28 eval). The crossing of  $\tau_0=0.8$  at epoch 24 opens the gate;  $\lambda(t)$  ramps to 0.225 by epoch 27. Epoch-27 values were read from the live run log.

## 421 D The Decay-Spiral Failure and Its Repair

422 Failure. In a pilot 50-epoch 0.5B run (v3), the eval-aware controller issued eval-driven  
 423 decays every two epochs for five consecutive episodes while held-out loss diverged, driving  
 424 the cumulative LR multiplier along  $1.0 \rightarrow 0.49 \rightarrow 0.34 \rightarrow 0.24 \rightarrow 0.17$ , breaching the  
 425 controller’s own nominal minimum (0.3) because the eval-driven pathway applied decays  
 426 without consulting the instability path’s floor. Memorization was choked: likelihood recall  
 427 stayed pinned near 0.02 and the gate threshold became unreachable. The run was stopped  
 428 to avoid burning compute on dynamics that the controller itself had frozen.

429 Diagnosis. The failure is structural, not incidental: in the memorization regime, held-out  
 430 divergence is the expected signature that the saturation gate waits for, so an eval-driven  
 431 damping rule without a floor is guaranteed to eventually fight the gate.

432 Repair. The eval-driven pathway now clamps the cumulative multiplier to  $m_{\text{min}}=0.5$  (Eq. 5)  
 433 and refuses further eval-driven decays at the floor. Validation proceeded in three stages: (i)  
 434 unit tests on a stubbed trainer assert the floor is respected exactly ( $m$  lands on 0.5, never  
 435 below; optimizer LR’s match base  $\times$  floor) among 138 checks; (ii) a dedicated 5-epoch GPU  
 436 run with an unreachable gate reproduced the failing regime ( $\lambda=0$  throughout, eval rising)  
 437 and showed one eval-driven decay ( $1.0 \rightarrow 0.7$ ) with the floor never breached; (iii) both  
 438 production runs in this paper clamp at 0.5 and hold (Fig. 4c) while memorization proceeds  
 439 unimpeded to saturation at 7B.

## 440 E Artifact Map and Log Reconstruction

441 Hugging Face Hub (user codewithdark). mlpr-qwen2.5-0.5b-instruct-50ep-adaptive-v4: fi-  
 442 nal adapter, probe head (probe\_head.pt), tokenizer, and full telemetry (gate\_history.json,  
 443 adaptive\_history.json, trainer\_state.json with 1,925 log entries, training\_log.txt, 1.5 MB).  
 444 The 7B repository mlpr-qwen2.5-7b-instruct-50ep-adaptive-v4 holds the epoch-26 adapter  
 445 (the trainer’s last in-progress hub commit) and the training log through epoch 26; its per-  
 446 epoch gate and controller records for epochs 1–25 are parsed from the log’s structured [MEM  
 447 GATE] / [ADAPTIVE] lines, and the epoch-27 record was captured verbatim from the live  
 448 log stream before termination. Weights & Biases (project knowing\_using\_gap): runs mlpr-  
 449 qwen2.5-0.5b-50ep-adaptive-v4 and mlpr-qwen2.5-7b-50ep-adaptive-v4 with per-step losses,  
 450 learning rates, gradient norms, per-epoch adaptive decisions, LoRA/probe matrix panels  
 451 every 5 epochs (168 heatmaps each of  $B$ ,  $A$ , and  $\Delta W$ ), and locked checkpoint artifacts.  
 452 Codebase: the released zip contains the full training stack (src/ with probe, LoRA setup,  
 453 gate callback, adaptive controller, likelihood evaluator, W&B matrix logger), the Modal  
 454 deployment (modal\_train.py), the dataset generator, the analysis and figure scripts, and  
 455 this paper under paper/.

## 456 F The Likelihood Recall Metric in Detail

457 For a memorization pair  $(x, y)$  the prompt is built by stripping the Answer: suffix from  
 458 the training text. The metric performs a single teacher-forced forward pass over prompt  $\oplus$   
 459 answer with left padding and records (i) strict first-answer-token exact match (3), and (ii) the  
 460 fraction of answer tokens whose greedy next-token prediction matches the reference (token  
 461 recall). Both are macro-averaged over the subsample. Compared with free-generation EM,  
 462 this removes decoding length, format, and repetition effects; the residual gap between token  
 463 recall and strict EM isolates single-token identity from multi-token assembly. Substring  
 464 containment of the target in the free generation complements both by detecting stored but  
 465 misframed answers—the 76.5% vs. 2% contrast in Table 1.